



Apparently the government's decision that the 2010 Commonwealth Games will be broadcast only in high-definition has galvanised the push toward HDTV in India. As a result, the state-owned broadcaster Doordarshan, which is expected to launch HD-TV on an experimental basis soon, has stated it will produce content for the Commonwealth Games in this format.

Cabling - HDMI

Although most flat panel TVs and media players and audio amplifiers still sport an array of different connectors for interfacing with other equipment, the High-Definition Multimedia Interface (HDMI) which started appearing on products in 2003, has become the default for hooking together digital audio and video equipment. One of its design aims was to minimise the tangle of wires connecting the various units of an AV system together by combining video, audio and control cables into a single cable with just one connector on either end. In this way cable clutter is minimised and the various units in a system can be chained together with just a single cable running between each one.

The high end audio market is notorious for taking things to extremes in the supposed pursuit of ever better sound reproduction. Exotic materials, specialist construction techniques and bizarre theories are regularly extolled as providing better sound, much of the time merely as an excuse for charging huge sums of money. Audio cables are a particular subject of controversy in this respect, even though these are only required to carry relatively low frequencies up to 20 KHz or so and it should not be particularly challenging to engineer a cable to carry these low frequencies over short distances.

HDMI is based on and is backwards compatible with, DVI-D and DVI-I and HDMI cables and signals can be connected to DVI devices using an adaptor. However not all of the HDMI functionality will be supported.

The HDMI specification is currently up to version 1.3 which adds support to the preceding 1.2 specification. Unfortunately, a number of current Blu-ray players, amplifiers and so on only have support for 1.2. To confuse things further, some features of the different specification revisions are optional features. For example, 1.3 adds optional support for output of Dolby TrueHD and DTS-HD Master Audio streams for external decoding. TrueHD and DTS-HD are lossless audio codec formats used on Blu-ray Discs.

HDTV image formats and re-scaling

There are two main HDTV image formats in common use: 1080i and 720p. These numbers refer to vertical resolution; the number of lines in the image from top to bottom. For an image with a 16:9 aspect ratio this means a 1080i image is 1920 by 1080 pixels and a 720p image is 1280 by 720. So the 1080i format would seem to have higher resolution.

Obviously for a display to be able to present 1080i or 720p signals its physical resolution must at a minimum match that of these signals. Looking at the specifications for HD Ready TVs can be a little confusing; for example, you can buy a 19-inch HD ready TV with a screen resolution of 1440 by 900.

HD Ready TVs use powerful image scaling algorithms and hardware to obtain the best fit for the variety of signal formats they must display. Because it was too expensive to design the early analogue TVs with a stable picture size that exactly filled the screen, the old TV signal standards allow for overscan; a border around the edge of the image that does not contain any important scene elements, such as title text. Analogue TVs are set up so that most of this overscan area overlaps past the edges of the physical display area. Digital TVs do not need to overscan since they can map images precisely to all the available pixels, but nevertheless they may allow some overscan in an attempt to provide the best display for signals using the earlier formats.

Video compression - MPEG-2, H.264

Raw video recording requires huge amounts of storage capacity. Because of this, all digital video is heavily compressed. Depending on the application, different compression schemes are used. For example, movies encoded to DVD use MPEG-2 (Motion Picture

Experts Group 2) while HDTV uses H.264. H.264 is equivalent to MPEG-4 AVC (Advanced Video Coding). If you want to display HDTV on your computer you will need a graphics card that includes H.264 decoding. Nvidia include this capability in a suite of features it calls PureVideo, so any of its graphics card products that support PureVideo are suitable for HDTV display. ATI has a similar suite of HD features it calls Avivo.

Although these sophisticated compression algorithms solve the huge size and bandwidth problems of raw high resolution digital video, they do at times introduce visible artefacts, most often seen as blocky patches in the image.

High-bandwidth Digital Content Protection (HDCP)

When the specifications for high definition video and Blu-ray were being drafted the major content providers such as Time Warner and Sony, concerned about the degree of piracy with the old analogue formats, insisted that all high definition video hardware should incorporate some form of content protection. HDCP uses a public key, private key encryption system to exchange handshakes between every device in the reproduction chain. If the handshake fails then the encrypted signal is not passed across the link. Public keys may be placed on a revocation list that is included on new media. If during the handshake, a public key is exchanged that is found to be on the revocation list, the device supplying the key is no longer recognised.

Some critics of HDCP have said that reverse pressures from the hardware manufacturers, concerned that implementing HDCP encryption would add too much to the manufacturing costs, have resulted in a weak encryption system that is relatively easy to break.

It is possible to buy HDCP dongles or strippers on the 'grey market' that can be inserted in the signal path to, for example, convert a non-HDCP compliant monitor. However, because these devices break the integrity of the encrypted signal chain, they are subject to key revocation.

Viewing Distance and Screen Size

Our eyes have sharp focus in the centre of the field of view surrounded by our peripheral vision. This means that there is a cone of vision or field of view where objects will be seen most clearly. As a result, for a given screen size there is an optimum viewing distance. Since vision varies somewhat, the viewing angle used by different organisations to calculate the optimum viewing distance also varies. SMPTE (Society of Motion Picture and Television Engineers) recommends a 30 degree viewing angle, THX (Tomlinson Holman's eXperiment - see boxout) 36 degrees (preferred) to 26 degrees.

The simplest way to work out the ideal viewing distance is to consider that given the SMPTE recommended viewing angle the distance to the screen should be 1.9 times the screen width (taking the THX recommendations the distance to the screen should be between 2.2 and 1.54 the screen width). Given the Dolby recommendations for a 5.1 speaker set-up radius of between 2 and 4 metres, this means that a 40-inch (roughly 1m) screen is not too large even at a 2m viewing distance. At 4 meters a 2m plus screen can be accommodated. HD images will of course tolerate closer viewing before the individual image elements become visible than will standard definition images.

Making your own HD video

There are now a number of video cameras on the market that shoot in HD formats. These record to MiniDV tape, or even to SD memory. None of these have HDMI connectors but use Firewire for their digital connection. For the 'prosumer' there is the Canon XHA1 or the Sony HDR-FX1 recording to MiniDV.

For un-edited playback you can plug these cameras directly into your display, normally using composite or RGB outputs and inputs, and using the camera itself as the playback device. Editing of HD can be done by capturing the video onto hard disk and using editing software such as Adobe Premiere on a Windows PC or Final Cut Pro on an Apple. The latest versions of these programs support HD editing. ■

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